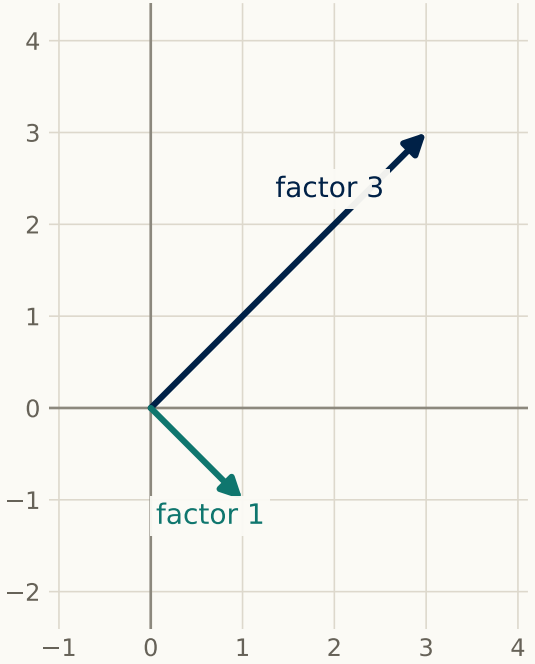
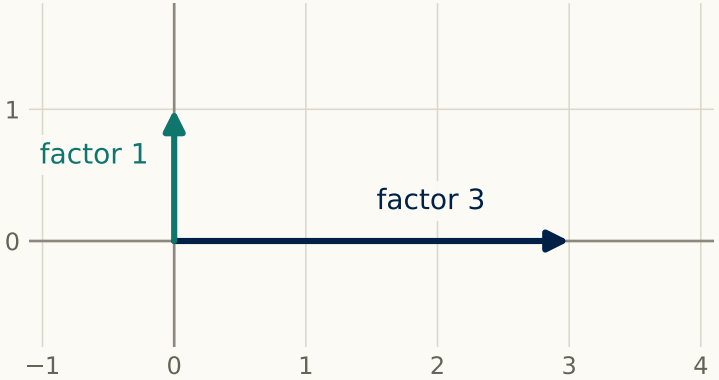
Eigenvector coordinates separate the stretches

basis (1,1), (1,−1)



coordinate matrix diag(3, 1)